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1984/1. Prove that
$0 \leq yz + zx + xy - 2xyz \leq \frac{7}{27}$
where x, y, z are non-negative real numbers for which
$x + y + z = 1$.

1984/2. Find one pair of positive integers a and b such that:
(i) $ab(a + b)$ is not divisible by 7;
(ii) $(a + b)^7 - a^7 - b^7$ is divisible by 7^7.
Justify your answer.

1984/3. In the plane two different points O and A are given. For each point X of the plane, denote by $\alpha(X)$ the measure of the angle between OA and OX in radians, counter-clockwise from OA $(0 \leq \alpha(X) < 2\pi)$.

Let $C(X)$ be the circle with center O and radius of length $OX + \alpha(X)/OX$.

Each point of the plane is colored by one of a finite number of colors. Prove that there exists a point Y for which $\alpha(Y) > 0$ such that its color appears on the circumference of the circle $C(Y)$.

1984/4. Let $ABCD$ be a convex quadrilateral such that the line CD is a tangent to the circle on AB as diameter. Prove that the line AB is a tangent to the circle on CD as diameter if and only if the lines BC and AD are parallel.

1984/5. Let d be the sum of the lengths of all the diagonals of a plane convex polygon with n vertices $(n > 3)$, and let p be its perimeter. Prove that

$$n - 3 < \frac{2d}{p} < \left[\frac{n}{2}\right]\left[\frac{n+1}{2}\right] - 2,$$

where $[x]$ denotes the greatest integer not exceeding x.

1984/6. Let a, b, c and d be odd integers such that $0 < a < b < c < d$ and $ad = bc$. Prove that if $a + d = 2^k$ and $b + c = 2^m$ for some integers k and m, then $a = 1$.